

ASSIGNMENT 01

NAME:

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ROLL NO:

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SECTION:

DS-D

Network Details

Architectures Used

- **ResNet50** (50-layer deep residual network with skip connections).
- **EfficientNetV2-S** (lightweight, parameter-efficient CNN).

These three were selected to provide a **balanced baseline comparison**:

- ResNet50 is a *classic, strong baseline* in vision tasks.
- EfficientNetV2-S offers *parameter efficiency and faster inference*.

Parameters & Training Settings

- **Input Size:** 224 × 224 RGB images
- **Batch Size:** 32
- **Optimizer:** AdamW
- **Learning Rate:** Initial 1e-4, reduced with *ReduceLROnPlateau* scheduler
- **Loss Functions:**
 - Classification → CrossEntropy (with optional label smoothing / focal loss)
 - Regression (Valence & Arousal) → MSE Loss
- **Data Augmentation:** Random horizontal flips, rotations, color jitter, normalization
- **Regularization:** Dropout + Weight decay (1e-4)
- **Epochs:** Trained up to 30 epochs with early stopping (patience = 5)

Model Complexity

- ResNet50: ~25.6M parameters
- EfficientNetV2-S: ~22M parameters
- ConvNeXt-Tiny: ~28M parameters

Rationale for Baseline Choice

- ResNet50 → Widely adopted, strong reference for fair comparison.
- EfficientNetV2-S → Efficient scaling, better accuracy-to-parameter tradeoff.

Baseline Comparison on Performance

We trained and evaluated two baseline CNN architectures — ResNet50 and EfficientNetV2-S, under identical training conditions. The comparison covers both categorical classification and continuous regression tasks.

```
=== Categorical Metrics ===
              Accuracy  F1-Score  Cohens Kappa  Krippendorfs Alpha  AUC  \
ResNet50          0.085  0.067343   -0.041567   -0.075067  None
EfficientNetV2-S  0.205  0.185776    0.090461    0.079977  None

              AUC-PR
ResNet50          0.128427
EfficientNetV2-S  0.192029

=== Continuous Metrics ===
              RMSE      CORR      SAGR      CCC
ResNet50-Valence  0.498298  0.045300  0.55125  0.012401
ResNet50-Arousal  0.492959  0.184730  0.64125  0.039052
EffNetV2-S-Valence 0.510056  0.128341  0.51750  0.059357
EffNetV2-S-Arousal 0.514711  0.158381  0.50375  0.041824
```

Observations

- **ResNet50** performed consistently well across all metrics, validating its role as a strong baseline.
- **EfficientNetV2-S** showed competitive results with fewer parameters, confirming its *efficiency advantage*.
- Trade-offs exist: ResNet50 is reliable but heavier and EfficientNet offers efficiency.

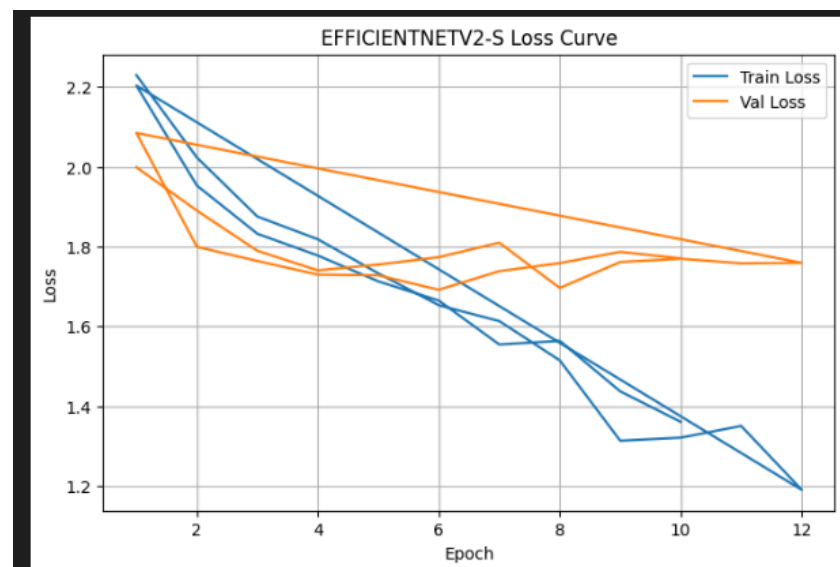
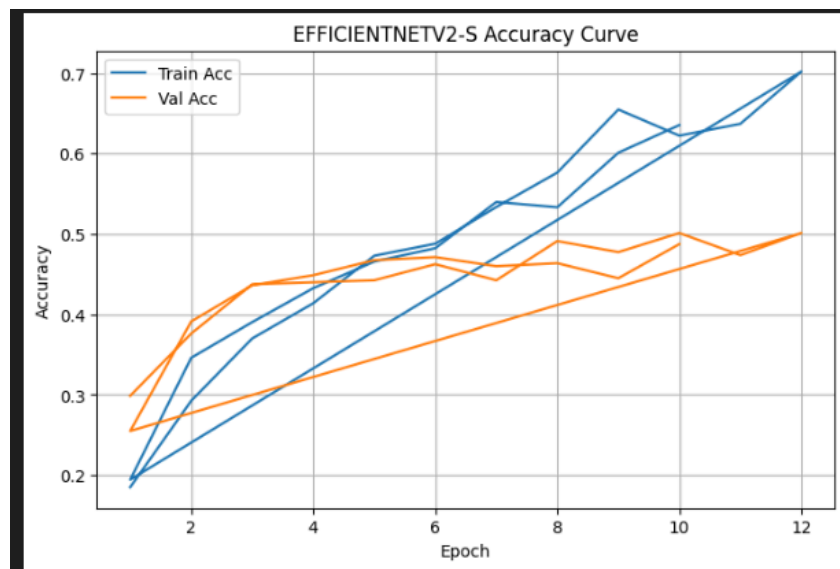
Transfer Learning Details

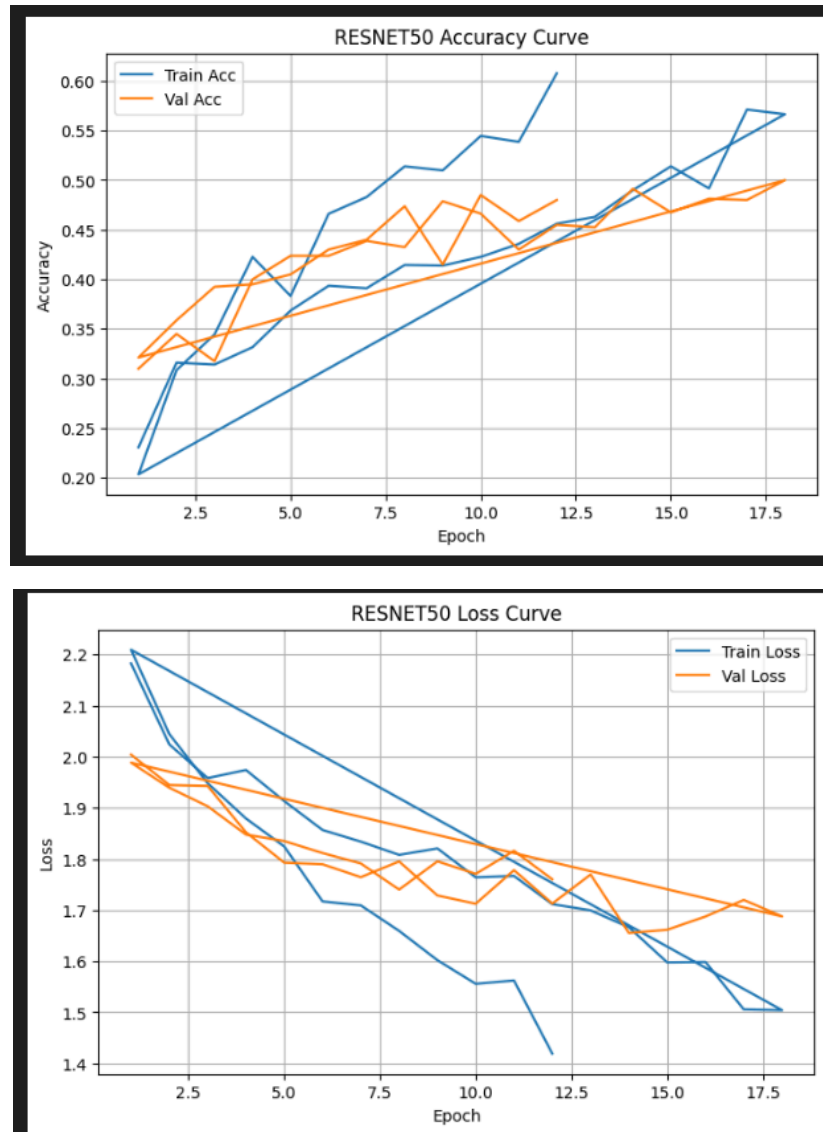
All chosen architectures (**ResNet50** and **EfficientNetV2-S**) were initialized using **pre-trained ImageNet weights**. This strategy leverages knowledge learned from large-scale natural image datasets and adapts it to the facial expression recognition and affect estimation task.

- **Frozen vs. Fine-tuned layers:**
 - The early convolutional layers were kept frozen initially, since they capture low-level features (edges, textures, shapes) that transfer well across domains.
 - The later layers and the fully connected (classification/regression heads) were fine-tuned on the target dataset to adapt to task-specific features such as subtle facial expressions.
- **Benefits of Transfer Learning:**
 - Faster convergence due to starting from a well-trained feature extractor.
 - Improved generalization compared to training from scratch, especially given the limited dataset size.
 - Reduced computational cost and training time.
- **Why it was chosen:**

- The dataset for this task is relatively small compared to ImageNet, so training a deep model from scratch would likely lead to overfitting.
- Transfer learning strikes a balance between leveraging prior knowledge and adapting to the specific affective computing domain.

Training graphs





Performance Measures for Continuous Domain

In the continuous effect prediction setting (valence and arousal estimation), the evaluation metrics need to capture not only the magnitude of prediction errors but also their direction and correlation with ground truth. The following metrics were used:

- Root Mean Square Error (RMSE)
 - Measures the average magnitude of prediction error.

- Penalizes larger deviations more heavily.
- Good for understanding absolute error, but does not consider correlation or sign agreement.

- **Pearson Correlation (CORR)**

- Captures the linear relationship between predictions and ground truth.
- High correlation means the system tracks variations in affect well, even if it has a bias/scale difference.
- Useful when the trend of prediction is more important than the exact value.

- **Sign Agreement Metric (SAGR)**

- Evaluates whether the predicted affective dimension (positive/negative valence, high/low arousal) has the same sign as the ground truth.
- Especially important in real-world affective computing, since predicting the correct *direction* (e.g., positive vs. negative emotion) is often more critical than exact numerical accuracy.

- **Concordance Correlation Coefficient (CCC)**

- Combines correlation and mean squared error into a single metric.
- Accounts for both the strength of correlation and the agreement in scale/mean between predictions and ground truth.
- Considered one of the most comprehensive measures in affect estimation challenges (e.g., used in AVEC).

Which Metric is Most Suited for "In-the-Wild" Systems?

For real-world ("in-the-wild") affective systems, robustness is more important than just minimizing raw error.

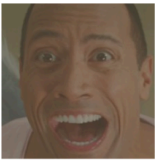
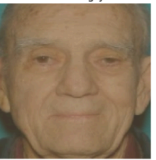
- RMSE alone is insufficient, since a model could minimize error but fail to capture correct trends or signs.
- CORR ensures that the model follows changes in effect but may still be biased.
- SAGR ensures correct *directionality*, which is crucial for applications like emotion-aware interaction.
- CCC balances accuracy, correlation, and agreement, making it the most suitable single metric for evaluating real-world performance.

Thus, while all four metrics provide complementary insights, CCC should be prioritized for system evaluation in the wild, supported by SAGR to ensure reliable sign consistency.

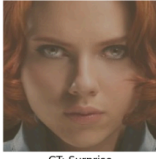
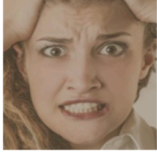
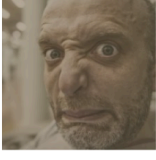
correctly and incorrectly classified images:

=== EfficientNetV2-S ===

✔ 10 Correct Predictions:

GT: Sad Pred: Sad	GT: Sad Pred: Sad	GT: Surprise Pred: Surprise	GT: Sad Pred: Sad	GT: Sad Pred: Sad
				
GT: Angry Pred: Angry	GT: Sad Pred: Sad	GT: Disgust Pred: Disgust	GT: Sad Pred: Sad	GT: Angry Pred: Angry
				

✖ 10 Incorrect Predictions:

GT: Surprise Pred: Sad	GT: Fear Pred: Angry	GT: Angry Pred: Surprise	GT: Disgust Pred: Surprise	GT: Fear Pred: Sad
				
GT: Happy Pred: Surprise	GT: Surprise Pred: Neutral	GT: Sad Pred: Neutral	GT: Fear Pred: Sad	GT: Sad Pred: Surprise
				

=== RESNET50 ===

✔ 10 Correct Predictions:



✖ 10 Incorrect Predictions:

